

Memory Game! (Team 18)

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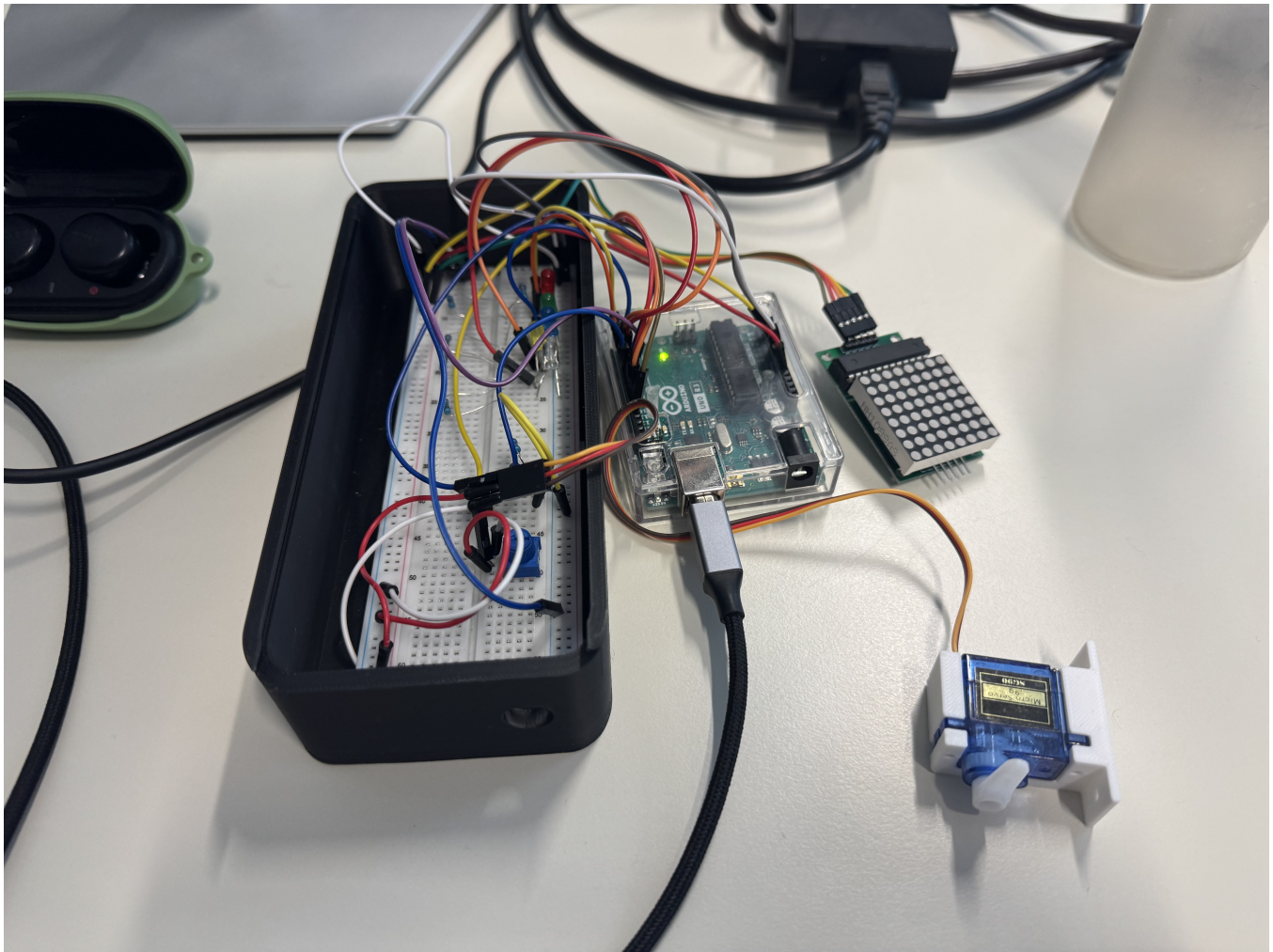


Figure 1: Overview of hardware part of the the memory game

ABSTRACT

Childhood is a crucial period for cognitive development, during which foundational skills such as attention and memory are formed. Among these, short-term memory plays an important role in retaining new information. However, children may not always have sufficient opportunities to actively train and strengthen their short-term memory in the world filled with distractions. Designing targeted practice games can effectively enhance their cognitive abilities by providing structured and playful ways to stimulate mental processes.

In this project, we aim to see if using some sort of a tool we can improve and enhance short-term memory in children users,

and how effective such a tool can be in both encouraging users to work on their memory skills, and ultimately in enhancing users' cognitive skills overall.

Our gamified tool relies on direct user interaction to play with the game and receive feedback. It also includes a score-tracking functionality to push users to enhance their memory skills and come back and play more. Since different users might be interested in enhancing different cognitive aspects, we provide games that focus on those aspects. We are utilizing an Arduino-based system that includes several interactive mini-games that involve a combination of visual, and auditory tasks. To make the tool more appealing, we

gamified the concept of a memory game to help users engage with our final product better.

At later stages of the project, we test the project with real users and gather feedback to enhance our tool's functionality, with a focus on its usability, and usefulness in a sense that how often users use our tool to practice memory enhancement instead of other activities they perform to spend time during rest time. We use experiments with participants to test our tool, and gather information via forms or verbally with short pseudo-interviews after the experiment to gather the information we require to make changes and evaluate how successful we were at achieving our goals.

1 INTRODUCTION

Human cognition is something which is often overlooked and under practiced. While people spend a lot of time developing other aspects such as physical (by going to the gym), mental (by performing meditation), etc., they often neglect cognitive aspects.

Working on cognitive skills and among them specifically memory skills can be an overwhelming job as it requires high degrees of attention and focus for long periods of time to be effective.

In this project, our goal is to create a usable tool which can be used by users to improve their memory skills. Rather than using classic methods, we use gamification to make the process of memory-improving skills much more desirable to users.

To make the game more dynamic, there will be different levels of complication. Every game starts with level one, and upon completion, the level increases. A higher level means a longer thing to be memorised.

In the rotation memory game, users will learn to focus their attention and improve their fine motor skills by precisely adjusting the potentiometer after observing the servo motor's movement. In the easiest levels, the servo will show a simple rotation, and the child will repeat it. In harder levels, the servo will show multiple angles in a sequence, and the child will have to remember and reproduce the full series, training memory, motor control, and concentration.

In the LED pattern game, users will train their visual memory and concentration. At first, simple patterns will appear slowly on the LEDs, and users will have enough time to memorize them. As the level increases, the number of lights will grow and the blinking will happen faster. User will need to remember more complex patterns with less time to observe, which will help develop stronger short-term memory and visual processing skills.

In the color counting game, the focus will be on fast visual processing and logical thinking. Using a set of 5 different color LEDs we show a sequence of LEDs lighting up, and ask user to replicate the pattern.

The sound recognition game will boost auditory memory and listening skills. At first, a simple single musical note will play, and the child will press the button matching that note. In harder levels, the MP3 module will play longer sequences with several notes, and the user will have to recognize the sounds in order and press the correct buttons. This will help user develop their ability to concentrate on auditory information, improving their memory and reaction to sound stimuli.

To support the players and help them learn how to play and improve, we will also develop a website. It will include explanations

for each mini-game and the parts that could not be developed for the game with Arduino set. This will make it easier for children, parents, and teachers to use the system effectively and make learning more fun.

We also have a life mechanism in which if the user makes 3 mistakes, they lose the game and have to start from the beginning, this makes trial and error and also guessing harder.

Overall, the project aims to create a fun, motivating, and educational experience that helps children gradually build important cognitive skills like reaction speed, memory, focus, hand-eye coordination, visual and auditory processing — all while playing in an engaging and supportive environment.

2 RELATED WORK

Santos [2] in a project created a minigame using a Teensy board. The mini-game is about remembering the turn-on pattern for a series of LEDs and reproducing the pattern. This is a similar mini-game to one of the mini-games we thought about. The project uses 4 different LEDs, and 4 buttons for each of those LEDs. For these components, at least 8 resistors are required, 1 per LED and button. The way this project works is that when the system is restarted, it shows a pattern of colorful LEDs, then the user has to press the corresponding buttons for each LED to replicate the pattern. A buzzer is used to give feedback to the user. The game starts with easy patterns and as the game goes on, the patterns become longer and more complicated. The focus of this project is on enhancing the cognitive skills of the user, but it only focuses on one dimension. We aim to focus on different dimensions, such as at least sound and visual memory, but the general idea of using a gamification method is the same, and very inspiring.

Music Box project [3] is an interactive device that integrates music playback with a variety of mini-games, aiming to create an engaging and educational experience. It is developed using an Arduino platform and incorporates components such as an LCD display, illuminated LED buttons, speakers, and potentiometers for volume adjustment. The housing is made from acrylic sheets, giving the device a durable appearance. The project offers four main modes: the Simon memory game, where players repeat sequences of lights and sounds; the reaction time game, which tests how quickly players can respond to visual cues; the free play mode, allowing users to create sounds freely; and the music mode, where preloaded melodies are played. At any point, players can press a dedicated white button to return to the main menu and choose another game mode. This approach is relevant to our memory game project, as both aim to enhance children's memory, reaction speed, and attention span through interactive gameplay. However, due to differences in the target age groups, the mini-games feature more challenging patterns in our project to better match the cognitive abilities of older children.

Authors of the paper "Design of a Set of Interfaces to Estimate Whether Computer Games Improve User's Skills and Abilities" [1] by Baines et.al., looked at how playing computer games can help people improve their reaction time, memory, and hand-eye coordination. They created three simple tests: one to measure how fast people react using an Arduino and a button, one to test memory by showing shapes and asking questions, and one to test hand-eye

coordination by throwing and catching a ball. After playing certain games, most participants showed better reaction times and memory, though the results for hand-eye coordination were mixed.

This study is closely connected to our project as we are also making an Arduino-based game system to help children improve reaction time, memory, and focus. Just like in their work, our project uses lights, sounds, and quick actions to train the brain in a fun way. Their results show that games like the ones we are designing can really help children get better at thinking fast and remembering things, which supports our idea of using simple, interactive mini-games for learning.

In our journey, all of these three project inspired us with our game design process, in a sense that how to properly give feedback to user using a gamified hardware. Furthermore, we got inspiration from [1]’s methodology to perform evaluation and to report our results.

3 USE CASES

Our game features three different cognitive skill enhancement, i.e., visual, auditory, and textual.

- **Cognitive skills, visual memory:** Our system for this game contains various buttons, LEDs, a servo-motor, and a potentiometer. Initially the user has to select a game (in this case, the angle memory game) by using the navigation buttons to go through the menu to pick the game. Note that for the prototype we do not use an LCD, and starting of the game is via long-pressing the reset-button for 1 second. After our system detects the long press, it flashes the green and then red LED once, indicating a new game is initialized. The servo then shows a random pattern of fixed angles to the user and blinks the green LED, indicating the start of the game.
- **Cognitive skills, auditory memory:** As for auditory memory, we offer two separate games. In both games, we play 7 different music notes, i.e., Do, Re, Mi, Fa, Sol, La, Si. In one game we ask the user to replicate the same note pattern, and in the other we ask the user to count the occurrences of a certain note. To make matters easier, we ask user to count the occurrences of the first note in the sequence. The user uses the web interface which looks like a piano, to interact with hardware. The piano metaphor was used to give user sense of interest in what their doing, and bring the abstraction of this game into a reality familiar to user.
- **Cognitive skills, textual memory:** For this use-case, we show user a sequence of numbers, e.g., 1529. We then ask them to repeat the sequence. In the web interface we provide a keypad similar to a phoner dialer. This uses the classic phone/dialing app on smart phone metaphor. It happens often that we want to dial a number but we have to check it multiple times. By practicing this game, and hopefully improving textual memory, user can better remember textual sequences.

In all of these games, if user makes a mistake, or gets promoted to next level, we show feedback using the 5 LEDs, by either playing a victory note by flashing all 5 LEDs three times, or for mistakes by flashing the red LED, and then re-playing the sequence to be replicated by user.

4 IMPLEMENTATION

In this section, we discuss the implementation process of our game. We also discuss the challenges and how to use the game.

4.1 Design

The first mini-game helps improve users’ visual cognitive skills. We hooked up a servo-motor, two LEDs, a potentiometer, and a button to our Arduino and created a game in which the user has to repeat the pattern of 3 randomly chosen angles by our system using the potentiometer to win the game. Upon success, the green LED turns and stays on, and with each correct part of the pattern after being registered by the button to our system, the green LED blinks indicating the correct part of the pattern is shown. If a wrong angle is submitted, the system blinks the red LED. After the game is done and also during the game at any time, user can long press the reset button to start a new game.

In the second game, we utilized a LED matrix that shows a sequence of number to user. The user has to then replicate the pattern using a keypad on the web interface. The arduino then sends the sequence to back-end using serial with baudrate 9600, back-end then gets the submitted sequence to front-end from user and compares these values. If incorrect, it signals both the hardware and the front-end that the result was wrong. The front-end then shows one less heart, and the hardware gives feedback to user that their answer was wrong by flashing the red LED, it then re-replicated the pattern for user. If the submitted value is correct, the back-end signals the hardware that it was correct, and the hardware shows a victory signal in which it flashes all LEDs 3 times, then hardware then shows another sequence with a higher length. If the user makes 3 mistakes at any point, they loose the game. If such a case happens, the hardware sends the highest completed sequence length to back-end and the front-end then shows this value.

For the third game, everything is the same, other than the fact that instead of an LED matrix showing numbers, 5 colorful LEDs with different colors blink to indicate a sequence. In the front-end then 5 buttons with these LED logos are shown to user that user uses to replicate the sequence. Upon success or failure, the same scenarios happen as with game 2.

In the fourth and fifth game, we utilize a passive buzzer, and by using PWM, we send signals corresponding to correct frequencies related to different music notes to user. In the fourth game, the user counts the number of times the first note played is repeated throughout the sequence. In game five, the whole sequence should be played. For both of these games, the web-ui offers a piano-style keyboard with which the user can submit the sequence. The piano metaphor was a nice fit for both of these games.

4.2 Challenges

Along the way of designing this system, we faced several challenges. The first challenge we faced was that none of our servo-motors

were able to be controlled with the ESP properly, i.e., the range of motion was limited to 90 degrees, as opposed to 180.

Another issue we faced was limited number of pins of the ESP. One idea that came to mind was using two ESPs and making them communicate using serial. But we needed serial for communicating with a back-end. We decided to use an Arduino UNO R3 instead of the provided ESPs to overcome this challenge. Using an Arduino solved the issue of not being able to properly control the servo-motor.

Using an arduino posed other challenges. The provided ESP had 4MB of flash, but Arduino UNO R3 only has 4KB of flash. This meant that the code we use had to be more efficient, and we could not freely use libraries for controlling our devices. After many refactoring cycles, we managed to make our code more efficient and deployable on the Arduino.

4.3 Usage

In order to use the system, the user initially sees the home page as indicated by Figure 2. This page allows users to enter any of the 5 games they want. Upon clicking on a game, they will be presented by one of the different UI's as indicated by Figure 3. In game 1, everything is done using the hardware. The user is shown a sequence of angles, and then has to replicate the angle using the potentiometer and submit the values one at a time using the button. If the value is correct, the next part of the sequence should be inputted by user. When the whole sequence is correctly given, the 5 LEDs shine and a new sequence is shown. This continues until user makes 3 mistakes and the game is over. For game 2-5, the user has to use the web UI to submit their results.

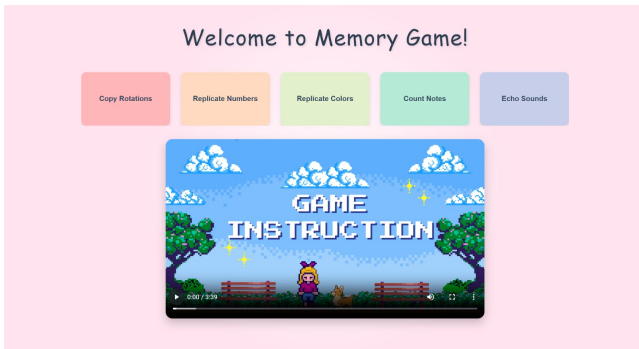


Figure 2: Home page of the system

The frontend is mainly developed using the Vue.js framework, combined with the Element UI component library to implement interactive features such as buttons and pop-up windows. In terms of web design, each game has its own individual page that utilizes CSS Grid Layout to clearly divide the interface into a game area and an instruction area, creating a user-friendly structure for easy operation.

The game's control logic operates as follows: After clicking the "Start" button, the game initiates and the Arduino module sequentially displays patterns. Users then input corresponding sequences by clicking buttons on the frontend and submit their answers. The

top-left corner dynamically displays remaining lives using a heart icon, while SignalR enables real-time reception of game feedback. The frontend synchronously updates the UI to show current error counts and remaining lives. The game automatically terminates when accumulated errors reach three times.

Our backend is built as an ASP NET Core Web API that exposes three REST endpoints - start, submit-sequence, and quit - all routed into one central InteractiveGameService. We also inject a SignalR hub so that, as soon as a guess is evaluated, we can push a "true" or "false" result in real time to any connected browser clients. Inside InteractiveGameService, we manage a single SerialPort connection to the Arduino and a small SequenceGameState<T> for each game (tracking the latest token sequence, remaining lives, and a "pending" slot for user input). When you hit start, we open (or reuse) the port, flush any old data, wait briefly for the Arduino to reset, and send "start N

r
n". A background Task.Run loop then continuously reads new sequences from the board, stores them, and waits for your HTTP POST to fill PendingSequence.

When you POST your guess, SubmitSequence copies it into PendingSequence, parses and checks it, writes back "true"/"false" to the Arduino, and broadcasts that same result over SignalR. A single break from the inner loop moves us back to read the next sequence. Key challenges included taming the Arduino's auto-reset (requiring buffer flushes and startup delays), handling inconsistent line endings to avoid read timeouts, and synchronizing a fire-and-forget background loop with incoming HTTP requests so nothing ever deadlocks or drops data.

5 PSEUDO-EVALUATION AND RESULTS

In this section, we go over the process with which we evaluated our system. In the first section, we discuss the study design, the questions, and how we tried to find answers to such questions. In the next section, we present the results. In the last section we discuss the limitation of the evaluation method we used.

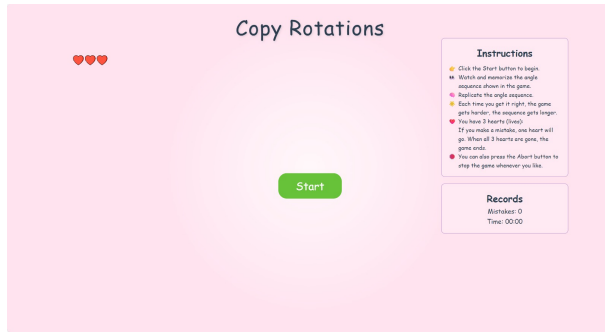
5.1 Study Design

To evaluate our tool, we want to answer the following questions:

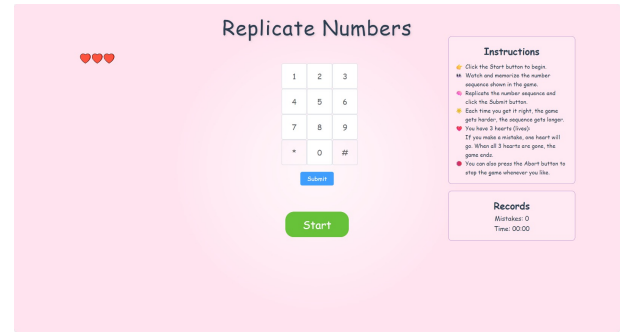
- **RQ1.** How effective is the tool in enhancing users' cognitive skills?
- **RQ2.** Do participants prefer to use the tool in their free time as a fun activity? (How effective was the gamification?)

We perform in-person experiments with other students, collect their performance data and ask these questions directly from them after the experiment and receive feedback from them. We then follow-up the experiment with a quick survey to get feedback regarding the system. We decided not to use interviews as the process would have been more time consuming for our unpaid participants, and also since the fact that completing a survey is much less demanding for them in this case.

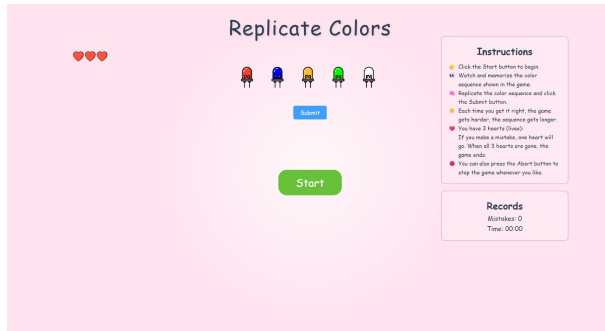
We use a performance Record for RQ1 To evaluate we will perform three experiments. First to introduce the user to the system and two others to check if they are improving after repeating exercise. To measure it we will record the highest record they achieve.



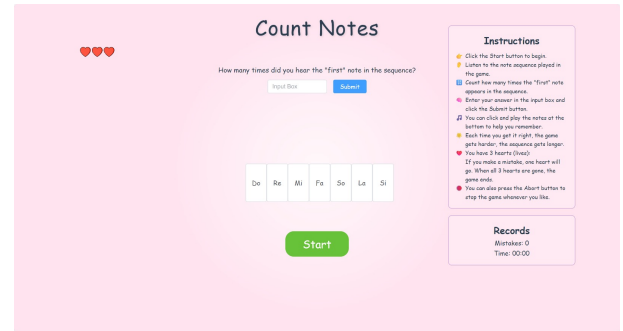
(a) Hardware handles the interaction for game 1.



(b) Interface for Game 2. A keypad is used as metaphor for phone.



(c) Interface for 3rd game. LEDs represent LEDs in hardware.



(d) Interface for 4-5th game. A piano is used as metaphor.

Figure 3: All interfaces for different games we implemented.

We use a Likert Scale for RQ2 To evaluate our system. Our Likert scale provided specific questions to users so that users can assess the experience and based on it we will make a general conclusion how likely they will return to the game.

Originally our project target audience was children, however, to test project with them and include them in the evaluation process is not possible, as we don't have access to children and it also imposes certain legal restrictions on us because they are underage. Therefore, for the project within this course, we will conduct an assessment among computer science students.

5.2 Results

In order to test our results, we explained the games and let our participants chose whichever game they want to perform the experiment. Ideally we would test all games, or have a larger user pool, but for the sake of this course, we decided to just evaluate the overall system as the general structure and idea of the games are the same, but there are minor differences between the user interface between different games. This means that participants can evaluate the whole system by testing a part of it, a part that they find more related to them, and more interesting to them.

Table 1 summarizes the results for answering RQ1. Although more tests should be conducted with a wider range users to be able to generalize these results, in game 1 effectiveness is not evident. The user was not able to learn using the device very easily, and was often confused as to which state they are in. The participant suggested using auditory feedback alongside the visual LEDs to

improve the usability of the game. In games 2-4, either reduction in time to complete, or enhancement of records is evident after some practice. For all the games, participants suggested more and perhaps better forms of feedback, such as haptic feedback to be utilized. Also Participant 2 suggested fixing the ordering of buttons for LEDs in web-UI for easing user interaction.

Table 2 summarizes the results for answering RQ2. Game 1 was perceived to be hard to learn, not easy to use, and not that interesting, but Participant 1 acknowledged that with better feedback the game could have a lot of potential. Game 2 and 3 were perceived positively, and believed to be easy to use and fun to play for user. They indicated they are the games they would come back to. Game 4 was also not positively perceived by Participant 2. They believed the game was a bit confusing and hard to learn. They believed by having better form of instructions the game would become viable.

We deliberately did not test Game 5 since it required some music-theory knowledge as user was required to know the music notes by heart.

We strongly believe by incorporating the feedback from users, and adding some quality of life features such as a better user instruction with a video on the front page, and also features for keeping user records and showing performance graph to them alongside a leaderboard, this game would a something users would come back to.

Table 1: Performance and Errors Across Three Attempts for Each Participant

Participant	Game	Try 1 (time-record)	Try 2*	Try 3*
Participant 1	1	2:00-4	2:03-4	1:44-4
Participant 1	2	2:23-7	3:19-10	1:56-6
Participant 2	3	3:04-7	3:54-9	2:45-6
Participant 2	4	2:07-5	1:45-5	1:37-5

Table 2: Participant Responses to Game Experience Questionnaire

Question	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
It was easy to understand how the game works.	P1-G1, P2-G4	P2-G2		P1-G2	
It was fun to solve the game task.			P1-G1, P1-G2		P2-G3, P2-G4
I felt stressed during the game.	P2-G3	P2-G4		P1-G2	P1-G1
I would use the game to practice my cognitive skills.		P1-G1	P1-G2, P2-G4	P2-G3	

5.3 Limitations

As with any qualitative work, our results are not generalizeable to all users. We only had access to students, and even there, the process was not funded, meaning we could not compensate our users. Another important limitation is that we built a large system, but evaluated parts of it with each user. This could mean that the results should not be put together and analyzed, and that a much larger group was needed for the test. But as mentioned before, the overall idea behind the games was the same, and the UI used to interact with the games was also very similar between different games, hence it was not necessarily required to test the whole system with every user, risking the addition of other challenges such as effect of fatigue during the test and between games, or other possible issues.

6 CONCLUSION

Cognitive aspects are very important, and often not exercised to become stronger. In this project, we designed a gamified usable

device with which users can practice and enhance their memory skills. We performed a user study to evaluate our system, and found pathways for improving the system even further.

6.1 Acknowledgments

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REFERENCES

[1] Christians Baines and Emanuele Lindo Secco. 2022. Design of a set of interfaces to estimate whether computer games improve user’s skills and abilities. *Acta Scientific Computer Sciences*.
[2] Rui Santos. 2014. Arduino simple memory game. Retrieved July 26, 2025 from <https://www.instructables.com/Arduino-Simple-Memory-Game/>.
[3] Yuvalt. 2023. Arduino music and game box. Retrieved July 26, 2025 from <https://www.instructables.com/Arduino-Music-and-Game-Box/>.